

Does a CNN Rely More on Global Shape or Local Texture?

A cue-conflict study with a fine-tuned ResNet-50 on a 4-class animal dataset

Code: github.com/660610873-Pollapat/DeepLearning-CNN

Research Question

When global shape and local texture provide conflicting class cues, which cue does our trained ResNet-50 rely on more during classification?

Prediction (written before the main cue-conflict experiment)

We predicted that the trained ResNet-50 would rely more strongly on local texture than global shape under the cue-conflict setup. This prediction was fixed before the main cue-conflict experiment and was not revised after seeing the experimental results.

Dataset

Animal Computer Vision Clean Dataset and Code (Emirhan Bulut, Kaggle). 4,000 images, perfectly balanced across 4 classes (Buffalo, Elephant, Rhino, Zebra; 1,000 each), verified by direct file count. Link: kaggle.com/datasets/emirhanai/animal-computer-vision-clean-dataset-code-cnnai. License: **CC BY 4.0** (Creative Commons Attribution 4.0 International), verified directly on the Kaggle dataset page. Attribution: dataset created by Emirhan Bulut, used here under the terms of CC BY 4.0.

Method Overview

- **Split:** 70/15/15 per class, fixed seed (700/150/150 images per class for train/validation/test).
- **Model:** ImageNet-pretrained ResNet-50, fine-tuned with only layer4 and the final classifier head unfrozen; the rest of the network stays at its pretrained weights.
- **Checkpoint selection:** best checkpoint chosen using the validation set only. The test set is never touched until after the checkpoint is frozen.
- **Cue-conflict stimuli:** Gatys-style neural style transfer (VGG19 features, Gram-matrix style loss, LBFGS optimizer, 150 steps) is used to combine the *shape* of one class's photo (content image) with the *texture* of a different class's photo (style image), producing an image whose two cues point to two different classes.
- **Trial design:** all 12 ordered class pairs (shape $\neq$ texture). For each pair, 40 content images (from the held-out *test* split) are each combined with 3 style images (from the *train* split), giving $40 \times 3 = 120$ trials per direction and $12 \times 120 = \mathbf{1,440}$ trials total. Both content and style pools are drawn with a fixed random seed, not hand-picked for outcome.
- **Frozen-model protocol:** the classifier is never retrained or fine-tuned again after the baseline checkpoint is frozen — it is used purely for inference throughout style-transfer generation and cue-conflict evaluation.
- **Classification of each stylized image** as a *shape choice* (prediction matches the content/shape class), a *texture choice* (prediction matches the style/texture class), or *other* (neither).

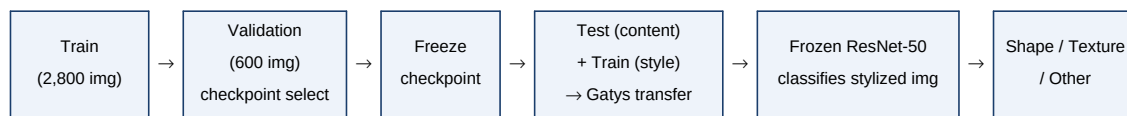


Figure 1. Experimental pipeline, left to right.

Non-independence note: the 1,440 trials are not 1,440 independent samples — each of the 40 content images per direction contributes 3 correlated trials (one per style reference). We therefore report results at both the *trial level* and the *content-image level* (3 style trials per content image averaged first, then averaged across the 40 content images); see Results.

Baseline: Can the Model Classify Normal Images First?

Before any cue-conflict interpretation is meaningful, the fine-tuned model must first classify ordinary, unmodified photographs well. We evaluated the frozen checkpoint once on the untouched 600-image test set (150/class).

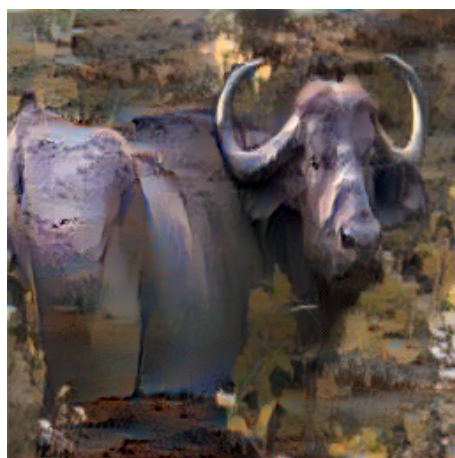
Metric	Value
Test accuracy	98.83%
Test macro-F1	98.83%
Buffalo (precision / recall / F1)	0.987 / 0.987 / 0.987
Elephant (precision / recall / F1)	0.993 / 1.000 / 0.997
Rhino (precision / recall / F1)	0.974 / 0.987 / 0.980
Zebra (precision / recall / F1)	1.000 / 0.980 / 0.990

Confusion matrix (rows = true, columns = predicted; order Buffalo, Elephant, Rhino, Zebra): [148,0,2,0] / [0,150,0,0] / [1,1,148,0] / [1,0,2,147].

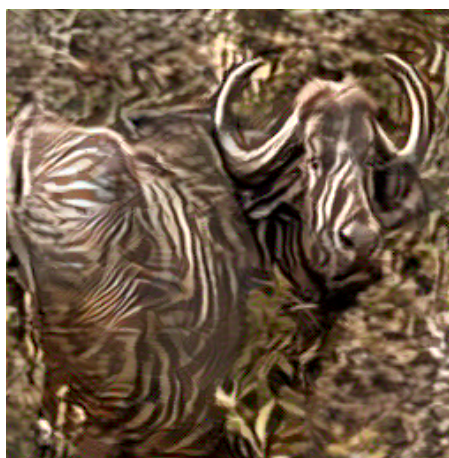
Assessment: every per-class F1 ≥ 0.98 with a near-diagonal confusion matrix — the model classifies all four classes well on natural images. This provides a strong baseline for interpreting the cue-conflict results below as the behavior of a well-performing classifier, rather than an artifact of poor natural-image classification.

Cue-Conflict Stimuli

The figure below shows two example stylized images: a typical direction (Buffalo shape + Elephant texture) and one of the three Zebra-as-texture directions discussed in Results, where the stripe texture is visually much stronger.



(a) shape=Buffalo, texture=Elephant — predicted Buffalo (shape choice, typical direction)



(b) shape=Buffalo, texture=Zebra — one of the low-shape-choice directions (68.3% shape here, vs. 87.4% overall)

Figure 2. Two representative cue-conflict stimuli generated with Gatys-style style transfer. The full set contains one example from each of the 12 directions (in examples/).

Important caveat on “texture”: neural style transfer does not isolate texture with perfect purity — it can carry along color and some local structure from the style image. We therefore treat the style-transfer condition as constructing a texture-defined cue (a style-defined visual cue built to conflict with shape), not as a pure, physically isolated measurement of texture reliance. Results are interpreted accordingly throughout this report.

Main Result: Cue-Conflict Experiment

Across all 1,440 trials, the model predominantly selected the **shape**-defined class over the texture-defined class:

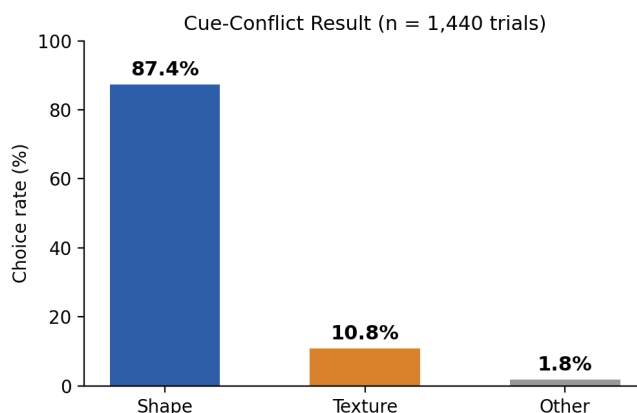


Figure 3. Trial-level Shape / Texture / Other choice rates, $n = 1,440$. Content-image-level aggregation (3 style references per content image averaged first, then across 40 content images per direction) gives essentially identical aggregate rates (87.4% / 10.8% / 1.8%), suggesting that the overall result is not substantially changed by averaging across the three style references.

Result vs. prediction: the hypothesis was NOT supported. We predicted texture would dominate; instead, *under our cue-conflict experimental setup, the trained ResNet-50 selected the shape-defined class in 87.4% of trials*, versus 10.8% for the texture-defined class and 1.8% for neither (average confidence 0.922). We did not revise the prediction after seeing this result.

Not Uniform Across Class Pairs: the Zebra Exception

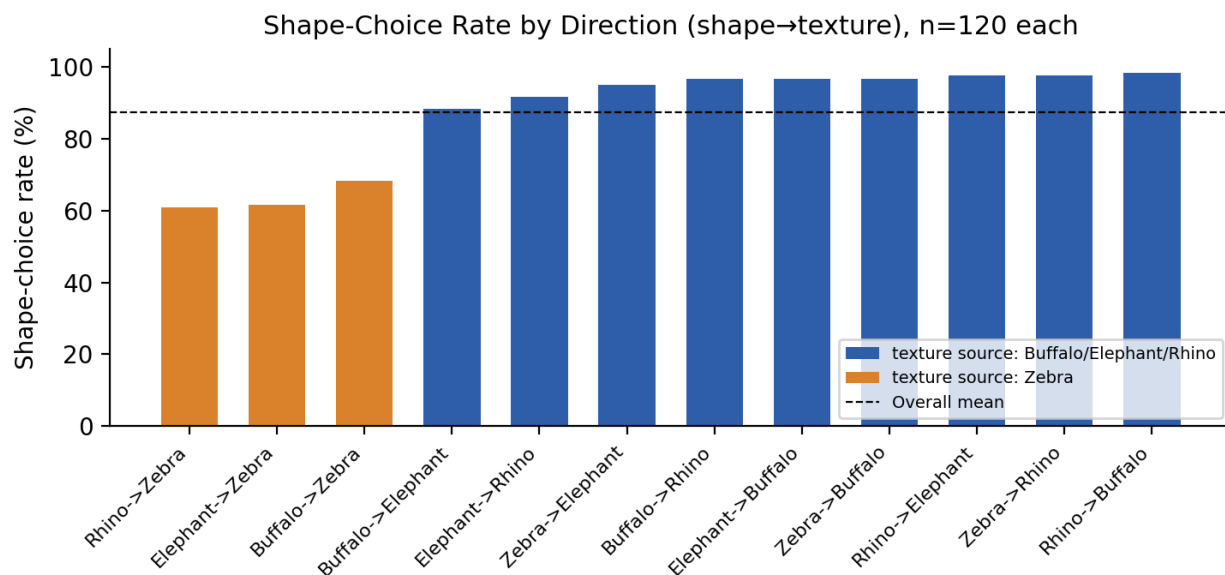
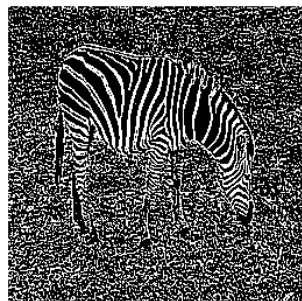


Figure 4. Shape-choice rate for each of the 12 directions ($n=120$ each), sorted ascending. Dashed line = overall mean (87.4%).

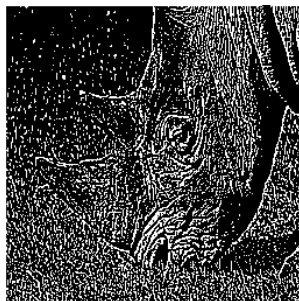
Although shape was dominant overall, the effect was not uniform across class pairs. In particular, the three directions using **Zebra as the texture source** (Buffalo→Zebra 68.3%, Elephant→Zebra 61.7%, Rhino→Zebra 60.8%) show substantially lower shape-choice than every other direction (88.3%–98.3%). By contrast, when Zebra supplies the *shape* instead (Zebra→Buffalo/Elephant/Rhino), shape-choice is back to normal-to-high (95.0%–96.7%). We present this as an observation rather than a firm causal claim: zebra stripes appear to be an unusually strong texture-defined cue specifically when acting as the style/texture source, but we did not run a follow-up experiment to isolate exactly why.

Supporting Control: Edge-Based Shape Representation

As a supporting check, we took the same 40 test-content images per class already used in the main experiment (160 images total, identical seeding) and converted each into an edge-based representation using PIL's `FIND_EDGES` filter followed by thresholding, substantially reducing color information while retaining intensity boundaries and local edges, then classified them with the same frozen checkpoint — no style transfer or retraining involved. Note that `FIND_EDGES` detects intensity transitions generally; it does not separate an object's outer contour from edges produced by internal texture (visible, for example, in the Zebra example below, where stripe edges remain prominent) — so this condition is an edge-based structural representation, not a clean isolation of shape with texture fully removed.



Zebra edge map (100% correct) — note stripe edges remain visible



Rhino edge map (37.5% correct)

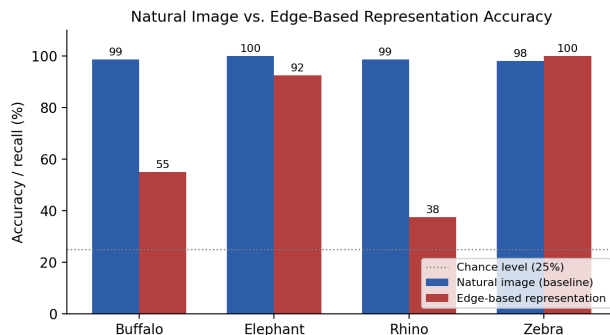


Figure 5. Baseline vs. edge-representation accuracy, overall and per class.

Overall: 71.3% accuracy (114/160), average confidence 0.581, versus 98.8% on natural images and 25% chance for 4 classes.
Per-class: Zebra 100%, Elephant 92.5%, Buffalo 55.0%, Rhino 37.5%.

Caveat (must be read together with the number above): this edge-based representation is out-of-distribution relative to the natural-image training data, so this result is treated as supporting evidence only, not as a second independent confirmation of equal strength to the cue-conflict experiment — the 27.6-point drop from baseline may reflect the distribution shift and the loss/transformation of multiple visual cues, rather than the removal of texture alone.

With that caveat in place, the edge-based representation still supports classification far above chance, which suggests that boundary and structural information carries substantial class-relevant information for this model. The per-class pattern is also informative: Zebra's edge map is classified perfectly even though its stripe edges remain visible in the representation, while Rhino and Buffalo edge maps are considerably harder to classify — how well this edge-based representation works as a cue is not uniform across classes.

Interpretation

We predicted the trained ResNet-50 would favor the texture-defined cue; instead it showed a **strong shape preference** under the cue-conflict setup (87.4% shape vs. 10.8% texture), corroborated by the edge-based representation still reaching 71.3% accuracy, far above chance. One plausible explanation is that this visually simple four-class task, combined with fine-tuning only the later layers (Layer 4 and the classifier head) of an ImageNet-pretrained model, may favor global outline cues such as body shape, horns, or ears. The result is not uniform, however — zebra stripes specifically pull shape-choice down to 61–68% when used as the texture source, suggesting that the Zebra style-defined cue can substantially influence the model when it acts as the texture source.

Limitations

- **Scope:** these findings describe this specific trained ResNet-50, on this dataset, with this training procedure, this style-transfer method, and this cue-conflict setup. They should not be generalized to “CNNs” as a class of models or to other architectures, datasets, or training regimes.
- **“Texture” is not perfectly isolated.** Gatys-style style transfer can carry along color and some local structure from the style image, so the texture condition is more accurately described as a texture-defined (style-defined) cue rather than a pure measurement of texture reliance.
- **1,440 trials are not 1,440 independent samples** — 120 trials per direction come from only 40 unique content images $\times$ 3 style images. We mitigated this by reporting both trial-level and content-image-level results (which agreed closely), but true independent-sample count is smaller than 1,440.
- **The edge-based control is out-of-distribution** relative to natural-image training data, and **FIND_EDGES does not cleanly isolate shape** — it detects intensity transitions generally, so edges from internal texture (e.g. zebra stripes) remain visible alongside the outer contour. This condition is therefore supporting evidence for the role of boundary/structural information, not a clean, confound-free second measurement of shape alone.
- **Only 4 classes, one dataset, one architecture (ResNet-50), one style-transfer method (Gatys/VGG19).** Whether the shape-preference finding holds for other class sets, architectures, or texture-generation methods is untested here.
- **Per-class asymmetry** (e.g. Rhino edge-representation accuracy 37.5% vs. Zebra 100%) shows reliance on this cue is not uniform across classes — the overall 87.4%/71.3% figures average over meaningfully different per-class behavior.

Reproducibility

All randomization (data split, content/style sampling, model initialization) uses a fixed seed (42), recorded in code. Pilot development images (used only to select the style-transfer method and step count) are drawn from an explicitly excluded index range and never appear in the 1,440 cue-conflict trials or the 160 edge-representation control images. Code, per-trial JSONL logs (`cue_conflict_trials.jsonl`, `silhouette_trials.jsonl`), and summary JSON files are included in the repository below so all reported numbers can be inspected or regenerated (the trained checkpoint itself, ~94 MB, is not included to keep the repository lightweight; re-running the training script with the same seed regenerates an equivalent one).

Code repository: github.com/660610873-Pollapat/DeepLearning-CNN

AI Usage Disclosure

AI assistance (Claude) was used in this project in the following ways:

- **Experimental design:** helped design and iteratively refine the cue-conflict protocol (dataset verification, train/validation/test data-role separation, seeded sampling, pilot-vs-experimental-data separation, trial/content-level non-independence handling, edge-representation control design).
- **Interpretation planning:** helped plan how to analyze and present results honestly, including scoping conclusions to this specific model/dataset/setup and flagging the texture-defined-cue and edge-representation-OOD caveats used throughout this report.
- **Writing assistance:** helped draft and structure this report from the experiment's own numeric outputs.
- **Where AI needed correction:** during pilot development, the AI initially misdiagnosed a noisy AdaIN-proxy output as a failed weight download, when the actual cause (identified by the student, who had written the pilot script) was an unregularized Adam pixel-space optimization. The AI's early framing was corrected before proceeding, and the project switched to a full Gatys optimization method as a result. All experimental code that produced the reported numbers (baseline training, main cue-conflict experiment, results analysis) was written and executed by the student; the silhouette control script was AI-written but reviewed and run by the student.